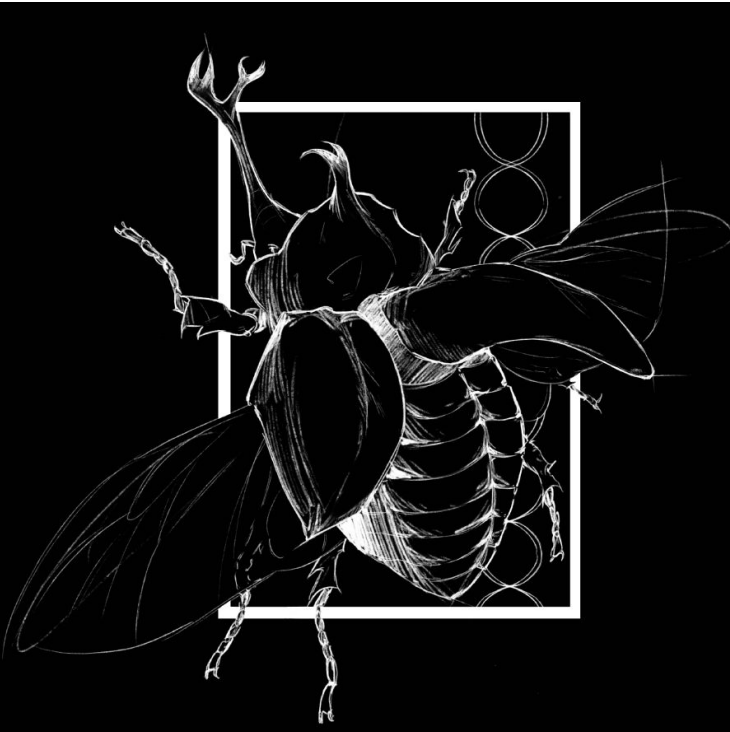




Background

With over 400,000 species, Coleoptera exhibits a staggering range of genomic and ecological diversity. While the ancestral beetle state is largely conserved, lineages frequently deviate through chromosomal fusion and fission events, which drastically reshape the karyotype landscape. While the ancestral beetle karyotype is hypothesized to be $2n = 20$, more recent lineages have variations resulting from fusions or fissions. Central to this study is whether these structural rearrangements are dictated by the amount of DNA content present. Similar research looks at why chromosome numbers and arrangement evolve the way they do. My project tests whether genome size and chromosome numbers correlate in beetles, providing deeper insight into the topic.

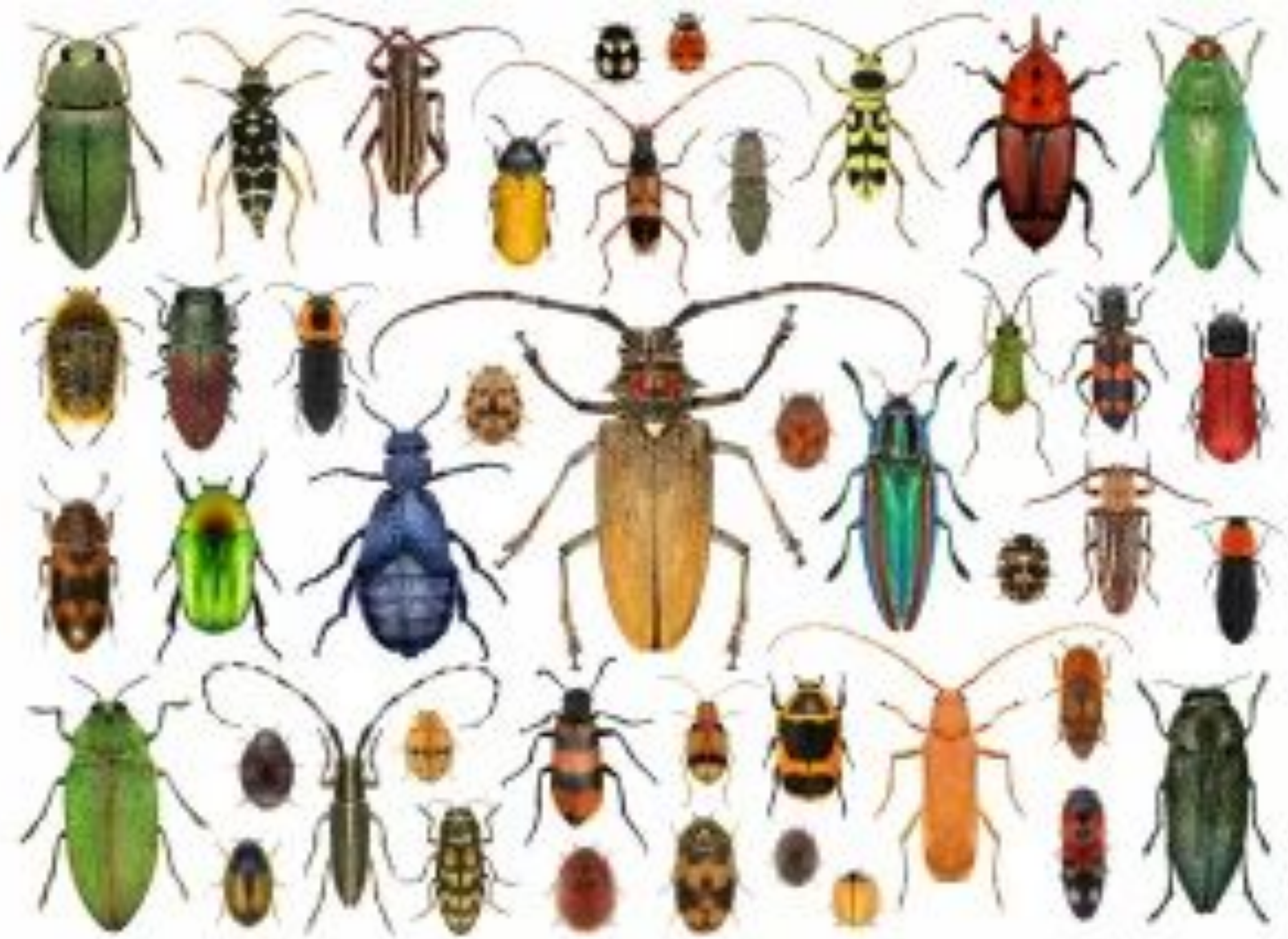
Study Design & AI

This study is a correlational evolutionary analysis that looks into whether genome size affects the rate of chromosome evolution. I studied around 300 species from Coleoptera and visualized the continuous evolution of genome size across the phylogenetic tree. By mapping genome size (C-value) as a continuous trait, we were able to account for the evolutionary history among species. This allowed us to treat genome size as a “discrete” state contingent upon the size of the genome in which those changes occurred.

AI assisted with phylogenetic setup, data organization, and R coding scripts. All methods are reproducible using open source R software.

Model Fitting

A Markov Model of Evoluton was fitted to this evolutionary model. A Markov Chain Monte Carlo (MCMC) statistical model was run for statistical estimations. This takes the sampling from the most likely area of the data, making the final results more robust and accurate. We set the model to explore the probability $n=3000$ times. To use a Markov model, the data was also discretized into 2 balanced groups: small and large genome sizes.



Results

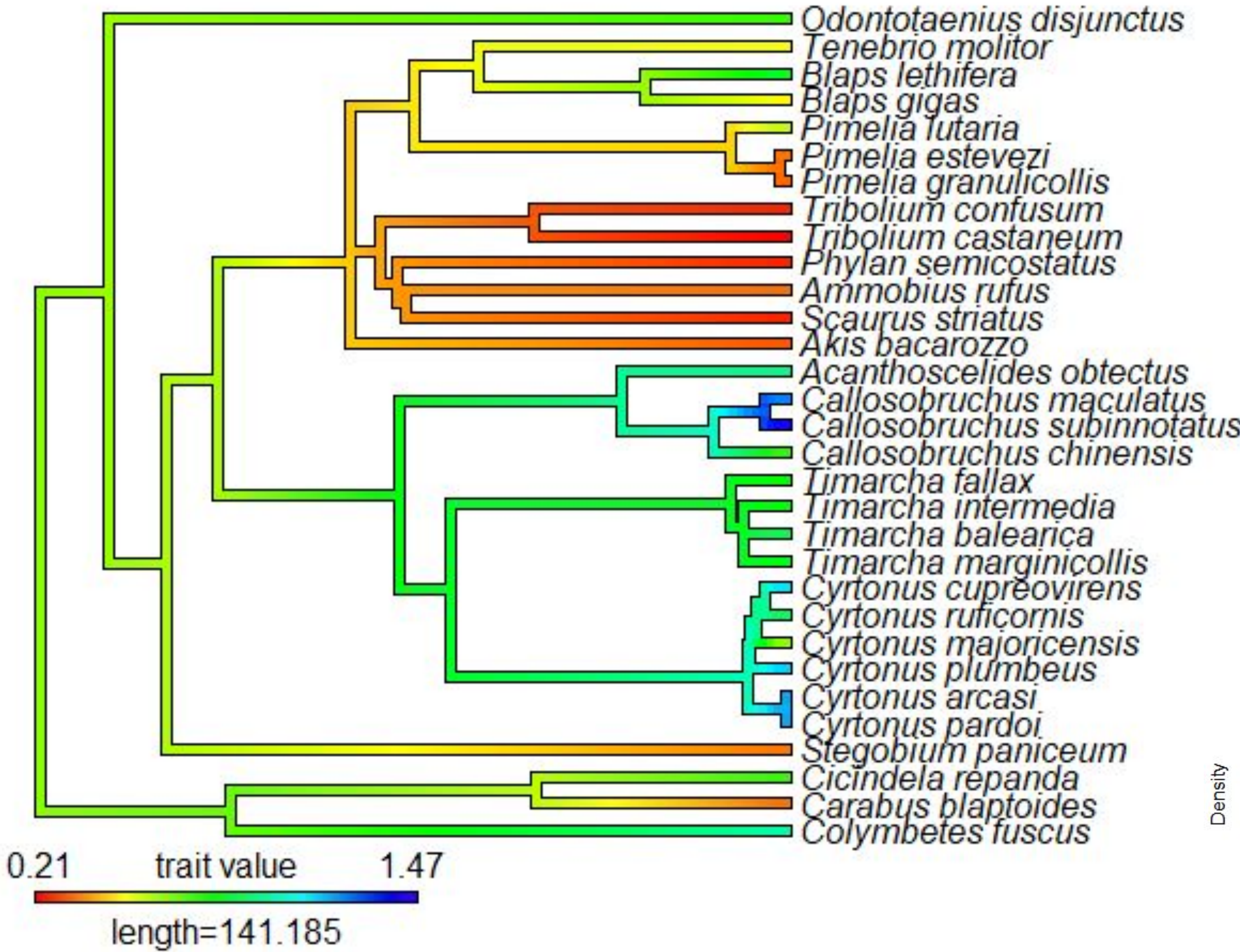
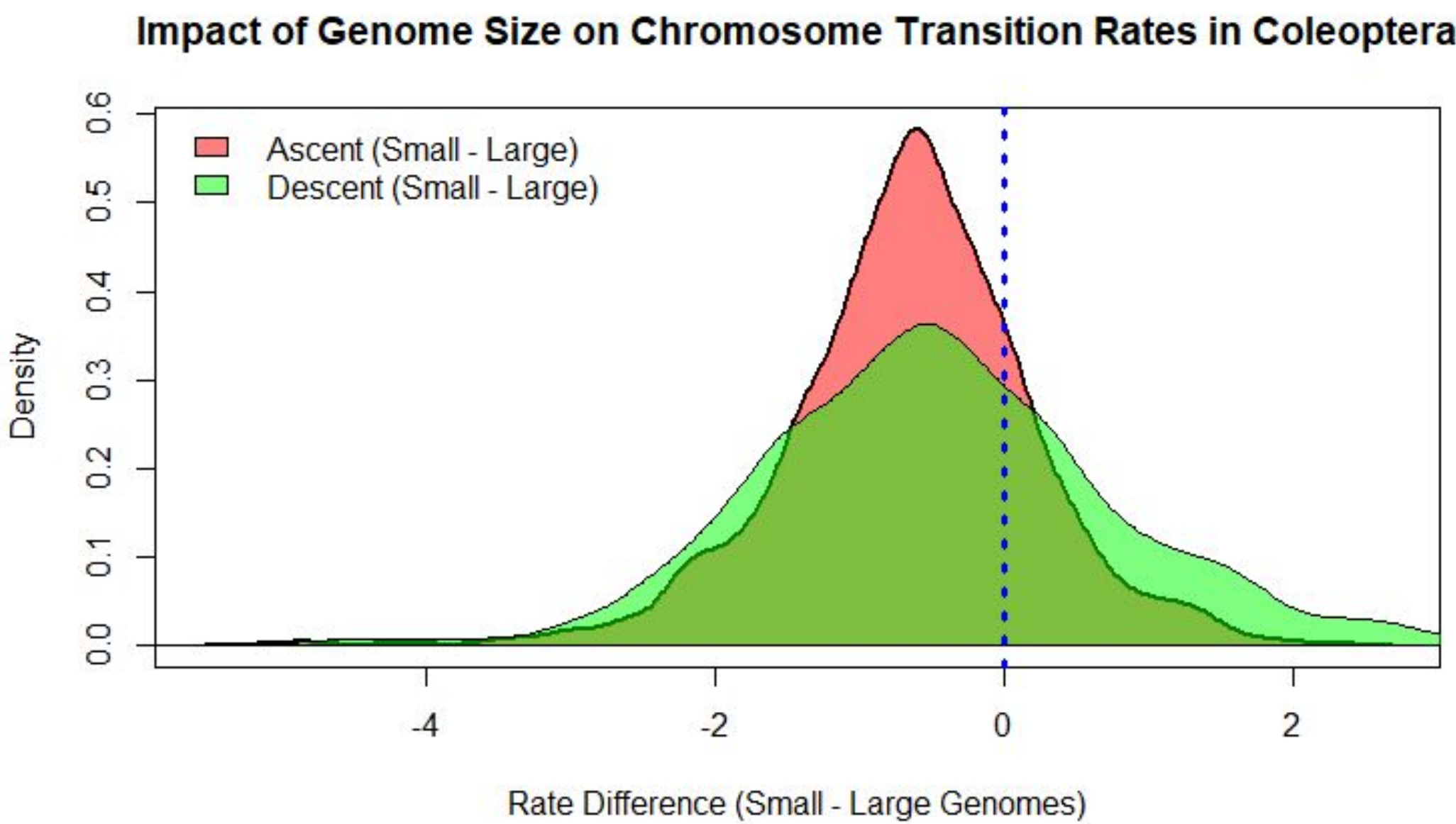


Figure 1. Evolutionary History of Genome size across Coleoptera. The color gradient shows the estimated ancestral genome size across lineages.

Figure 2. Impact of Genome Size on Chromosome Transition Rates. Density plots show the differences in evolutionary rates between “small” and “large” genome sizes. This threshold was measured by the median value.



Discussion

While our results do not support a robust, universal link between genome size and chromosomal evolution in Coleoptera, the high degree of variability observed across our dataset provides a compelling snapshot of genomic complexity. In this study, we assessed the C-value, the total DNA content within a haploid nucleus—to determine if the physical “bulk” of the genome serves as a driver for karyotypic change. Visualization of these traits across the phylogeny (Figure 1) confirms that genome size is largely conserved among closely related lineages, yet this conservation does not translate into a predictable pattern of chromosome change. Using a Bayesian MCMC framework, we found that the posterior distributions of rate differences did not shift entirely to one side of the spectrum (neither fully positive nor fully negative), rendering the results statistically inconclusive. However, we noted a suggestive directional trend: there was an 80% posterior probability that chromosome gain (ascent) occurs faster in large genomes, compared to a 67% probability for chromosome loss (descent). This disparity suggests that while genome size might slightly influence the “volatility” of a lineage, it is likely not the primary driver of evolution. A major limitation remains the relatively small sample size caused by the limited overlap between available genome size and chromosome datasets. Given that Coleoptera represents one of the most diverse groups of organisms on Earth, this high degree of variability suggests that chromosomal evolution may be dictated more by specific ecological pressures or life-history traits than by inherent genomic properties.

Conclusions & Acknowledgements

In summary, while our Bayesian analysis revealed suggestive directional trends, we cannot definitively state that genome size dictates the rate of chromosomal evolution across the Coleoptera. This research will continue in the upcoming semester, shifting focus toward resolving these patterns by significantly expanding the taxonomic sampling. Future work will bridge existing gaps between genome size and chromosomal datasets to improve the power of our Bayesian inferences. Furthermore, upcoming analyses will implement more rigorous statistical frameworks, such as treating C-values as continuous variables and testing for the influence of life-history traits. I would like to thank Dr. Heath Blackmon for their guidance and support throughout this project. This study utilized publicly available databases from the Animal Genome Size Database and the Blackmon Lab at Texas A&M University.